

Exploring Designers' Expectations From a Virtual Reality User Insight Tool: KitchXR

Okan Efe Öğretmen

oogretmen19@ku.edu.tr

KARMA Mixed Reality Lab, KUAR
Research Center, Koç University
Istanbul, Türkiye

Saliha Akbaş

sakbas19@ku.edu.tr

KARMA Mixed Reality Lab, KUAR
Research Center, Koç University
Istanbul, Türkiye

Asım Evren Yantaç

eyantac@ku.edu.tr

KARMA Mixed Reality Lab, KUAR
Research Center, Koç University
Istanbul, Türkiye

ABSTRACT

This paper investigates how designers envision using virtual reality as a spatial interaction tool for collecting user insights during product development. We developed KitchXR, a virtual kitchen prototype, to elicit expectations from 17 professionals in the home appliance industry, including UX/UI designers, industrial designers, and consumer insight specialists via an expert evaluation study. Through thematic analysis of semi-structured interviews, we identify key challenges in current UX research workflows, particularly limited access to user data and constraints on the timing of insight collection. Our findings highlight 3 opportunity areas for VR-based user insight tools: (1) enabling complex and context-rich testing scenarios, (2) integrating bio-sensory and behavioral data, and (3) supporting the revisitation of early design decisions. By centering designers as users of spatial interaction tools, this study contributes to emerging discourse at the intersection of extended reality, user research, and collaborative design workflows.

KEYWORDS

Virtual reality, user insight, consumer insight, virtual prototyping, user experience, expert evaluation study

ACM Reference Format:

Okan Efe Öğretmen, Saliha Akbaş, and Asım Evren Yantaç. 2025. Exploring Designers' Expectations From a Virtual Reality User Insight Tool: KitchXR. In *ACM Symposium on Spatial User Interaction (SUI '25)*, November 10–11, 2025, Montreal, QC, Canada. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3694907.3765914>

1 INTRODUCTION

Product design and development is a complex, multi-stage process that demands collaboration across diverse disciplines. Teams—including decision-makers, UX researchers, product designers, engineers, marketing, and finance—must stay aligned throughout, each bringing distinct priorities and working styles. From the earliest stages, consumer or user insights serve as a shared language, bridging disciplinary divides and informing key decisions. User Experience (UX) research has become central to driving innovation and producing meaningful outcomes in both product and service design.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SUI '25, November 10–11, 2025, Montreal, QC, Canada

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1259-3/2025/11.

<https://doi.org/10.1145/3694907.3765914>



Figure 1: The KitchXR virtual environment, designed as a modular, immersive testbed for UX research in home appliance design. Scene shows one of several customizable kitchen layouts, where participants interacted with smart appliances, UI elements, and spatial arrangements.

As Hassenzahl [21] emphasizes, UX goes beyond functionality and usability to encompass emotional and experiential qualities that shape users' relationships with products. By leveraging prototypes of varying fidelity, designers gather insights to guide decisions before production. Beyond feedback, UX research also links design stages and facilitates collaboration across teams.

Researchers use various techniques to understand users and envision user expectations in context. For example, Oliver et al. [35] replicated a real kitchen in a lab, outfitted with sensors and projectors, to study user interactions. The Living Labs methodology enables user-centered evaluations of technologies in real-life settings, particularly in smart homes, IoT, and ubiquitous computing. PlaceLab [23], for instance, allows researchers to explore ubiquitous technologies in authentic domestic environments. Similar methods have been applied in studies on assisted cooking [43], IoT infrastructures [19], and well-being research in kitchen labs [41]. While valuable for producing ecologically valid insights, Living Lab setups are often resource-intensive, time-consuming, and lack flexibility for rapid iteration.

Another approach might be using virtual platforms and tools to collect user insights. Although a wide array of methods (e.g., contextual inquiries, diary studies, and eye-tracking systems) and software platforms (e.g., Figma, Axure, ShapesXR, and NVIDIA Omniverse)

already support digital prototyping and user insight collection, there remains a strong industry demand for swift, user-friendly, and easily integrable tools that embed HCI research methodologies within Extended Reality (XR) environments. A Virtual Living Labs approach may offer new opportunities to investigate interaction patterns, environmental constraints, and behavioral routines in situ.

Several studies have explored the early adoption of XR technologies in industrial settings; Berg and Vance [4] illustrate how Virtual Reality (VR) facilitates early-stage design evaluations and spatial collaboration; Fillatreau et al. [15] demonstrate the use of immersive 3D checklists for training and procedural guidance; and Wang et al. [42] review AR/VR implementations in digital twin systems for product development. Alongside these developments, studies have been questioning whether physical prototypes can be meaningfully augmented or supplemented by immersive XR technologies in UX workflows [39, 44]. With early-adopter companies increasingly embedding XR across various stages of the design pipeline [4], and as social VR and Metaverse platforms become more accessible, new opportunities are emerging for immersive, distributed collaboration and prototyping. More recently, Marques et al. [33] examined how XR can support hybrid collaboration by enabling remote participants to engage in co-located design tasks. In VRception, Reipschläger et al. [20] investigate how VR can be used as a prototyping environment for cross-reality experiences, spanning the Virtuality Continuum states proposed by Milgram [38].

Despite this momentum, existing literature continues to prioritize engineering, simulation, or training applications, offering limited insight into the specific expectations, challenges, and workflows of UX researchers and designers. In this rapidly evolving landscape, understanding the designer's perspective is critical, not only to inform the development of XR-based insight tools but also to ensure their meaningful integration into real-world, cross-functional design environments, particularly during early conceptual and ideation stages. This paper addresses that gap by focusing on product designers and researchers as the primary users of VR-based insight tools. With an expert evaluation approach, it investigates their expectations, challenges, and envisioned roles for immersive environments such as KitchXR in enabling context-rich, iterative UX research within multidisciplinary design teams.

This paper explores the following research questions: (1) What challenges and limitations exist in current UX research workflows for home appliances? (2) How might immersive VR extend or enhance these workflows for iterative testing? To investigate these questions, we developed a VR kitchen prototype and conducted an expert evaluation study with 17 professionals from diverse design and research related roles, including UX/UI designers, industrial designers, and consumer insight specialists. Following participants' interactions with the prototype, we conducted semi-structured interviews and analyzed their feedback using thematic analysis. The VR application we developed (Fig. 1) served both as a probe for ideation and a testbed for evaluating designers' expectations for immersive UX research environments.

Our findings indicate that VR can address two major challenges in existing UX research workflows; limited access to user data (especially across distributed contexts) and throughout different stages of the design process. Participants identified several domains where

VR can enhance insight collection—particularly in evaluating practicality of use, material quality and aesthetics, and user-product interaction. From our analysis, three key design opportunities emerged for future VR-based UX tools: (1) enabling complex, scenario-based testing environments; (2) integrating bio-sensory data for deeper behavioral analysis; and (3) supporting the revisitation of early design decisions through iterative prototyping. These findings contribute to the growing body of HCI literature on the role of XR technologies in professional design practice, emphasizing a designer-centered approach to immersive user research.

2 RELATED WORK

While XR technologies have long been applied in visualization, rapid prototyping, and simulation, a growing body of work has begun to explore their value in UX research. These studies highlight XR's practical benefits for immersive design reviews, accelerating prototyping, and collecting feedback in context-rich environments. We review two key threads of prior work: (1) the use of XR across different design stages, and (2) the role of multimodal and context-rich immersive environments in UX research. Together, these threads provide the foundation for our investigation into designers' expectations, challenges, and practices when using XR-based user insight tools like *KitchXR*.

2.1 XR in Different Design Stages

This subsection explores how XR, particularly VR, is integrated into various phases of product development—from early ideation and collaboration to prototyping and testing—revealing opportunities and limitations in its application for user-centered design. Despite persistent limitations in generalizable hardware/software configurations and open-source availability, VR has been widely adopted in the industry to support decision making and collaborative workflows throughout the design process [4, 15, 42]. Numerous studies demonstrate how VR enables designers and engineers to explore, evaluate, and iterate on concepts in early stages, often reducing cost and development time [11, 13, 14, 22].

Providing a spatially realistic and context-rich experience can facilitate more accessible and grounded decision making [8]. Berg [3] presents a case study of industry professionals using VR for immersive design reviews, allowing stakeholders to evaluate 3D models in a virtual setting that mirrors real-world spatial arrangements. This supports clearer evaluation of design issues and more informed improvements. Similarly, Krauß et al. [26] document recurring bottlenecks in XR prototyping workflows, identifying challenges such as tool interoperability, time/resource intensity, and a lack of integration into broader UX research pipelines. VR's role in co-design has also received growing attention. Wang et al. [42] emphasize VR's ability to facilitate real-time remote collaboration among inter-professional teams and users. In a foundational survey, Berg et al. [4] identify widespread use of AR and VR by designers and engineers, primarily for visualization or collaborative design tasks [2, 15, 31]. These studies emphasize design professionals' practical needs during ideation and prototyping stages, including effective communication and spatial exploration [26]. However, while these tools are often evaluated through the lens of end-user experience,

few studies investigate designers' experiences using XR to collect user insights during the design process [37, 45].

Muller [34] demonstrates VR's role in conceptual-stage ideation, illustrating how immersive tools can address gaps not yet covered by AR-based methods. These findings suggest that XR tools hold potential not only for visual communication but also as user insight platforms throughout the design journey—offering behavioral, verbal, and even physiological data. Beyond VR, other XR modalities—particularly AR—have been used to support iterative design processes. Cascini [10] shows how AR enhances "co-creative" workflows, facilitating dialogue between designers and end-users. Knierim [25] uses mobile AR to explore user expectations in home environments, employing it as a probe for ideation and iteration. Marner et al. [32] develop a SAR prototype to support communication between clients and architects, suggesting that even in spatially constrained domains like kitchen design, XR can support collaborative envisioning. Grünefeld et al. [20] introduce the VRception toolkit, a system that enables developers to rapidly prototype and test cross-reality experiences entirely within VR. The toolkit supports multi-user collaboration, voice communication, and avatar representation. Based on findings from user studies, the authors suggest that such an approach facilitates the development of cross-reality concepts by enabling rapid iteration and testing, as well as both remote and co-located collaboration. In another study, Giunta's review of AR in industrial design maps prior work onto the six phases of the "generic design process model" by Pahl et al. [36]—from Task Definition to Product Documentation—identifying persistent gaps in AR's application for early and final stages [17].

Building on these insights, our study positions VR not simply as a tool for design presentation or co-creation, but as a tool for collecting user insights that inform iterative design decisions. In the *KitchXR* study, we examine how designers and UX professionals envision immersive environments for collecting data—be it behavioral feedback, task performance metrics, or reflections—to shape ongoing design. Unlike studies that evaluate user experience within VR products, our work asks: how can VR serve as a research environment for designers themselves? This shift repositions designers as critical users of immersive research tools.

2.2 Context-Rich and Multimodal Approaches to Immersive UX Research

This subsection reviews how immersive environments—particularly those using multimodal and spatial interaction—are used to simulate real-life usage scenarios, fostering empathy and providing richer data for design research. A growing body of research shows that VR supports empathy-building and perspective-taking, both essential to understanding user experience in context [12, 40]. These qualities are especially useful in domains where real-world testing is costly, dangerous, or logistically challenging—such as automotive design, kitchen appliances, or healthcare.

In the ARCAR project [16], users wear a mixed reality headset with eye-tracking and hand-tracking capabilities while participating in driving simulations. This setup allows researchers to capture real-time data on user attention, gestures, and perception under simulated environmental conditions. Similarly, VR-OOM [18] uses

Wizard-of-Oz techniques in VR to understand passenger experiences in autonomous vehicles—focusing on how users interact with interfaces in situ. Bordegoni et al. [5, 6] show how visual, haptic, and auditory modalities can be combined to simulate appliance usage, enabling realistic virtual prototyping. Kuliga et al. [27] compare walkthroughs of real and virtual buildings, finding that VR can approximate empirical research conditions by providing strong spatial presence and contextual realism. Frameworks such as the XR Safety Kit by Burova et al. [9] further advance this approach by combining gaze tracking, behavioral logging, and task progression metrics into a single virtual system. Similarly, Alce et al. [1] demonstrate how body-tracking tools and VR can be combined to prototype wearable AR systems—highlighting the role of embodied interaction in iterative design. Game-based environments also offer insight. Rooney et al. [30] present "Serious Gordon", a VR kitchen simulation used to teach hygiene standards to food industry workers. While their study suggests that real-world prototypes still hold advantages for procedural learning, others like Pettersson et al. [39] report no significant difference in user responses between field and VR-based evaluations. Hoggemueller et al. [22] further nuance this view, comparing 2D video, 360° VR, and fully immersive CG environments—each proving valuable for different UX research goals.

Collectively, these studies reveal the richness of multimodal and context-aware VR for simulating use scenarios and collecting layered user feedback. However, few explicitly examine how designers can make use of rich qualitative, quantitative and behavioural data collected through XR prototyping and insight collection tools. That is, they rarely ask how these tools serve designers' insight-generation needs, or how immersive platforms might be configured to support iterative workflows. Our work addresses this by repositioning designers—not users or engineers—as core users of XR tools, investigating how they interpret, evaluate, and act on the insights gathered in immersive contexts like *KitchXR*. This perspective reframes the design process itself as a form of inquiry, where XR serves both as medium and method.

3 METHODOLOGY

Here in this section, we explain the methodological approach in detail, study procedure we followed (illustrated in Fig. 2) and how we developed the VR prototype.

3.1 Design and Prototyping of the VR Kitchen Experience

We designed and developed a room-scale immersive VR application for the VR Kitchen experience using Unity. Scenes and assets were modeled in Blender, with its plugin named Archimesh, and the interactive elements were implemented in Unity. The application runs on a VR-capable laptop connected to an HTC Cosmos headset. The experience consists of four scenes (Figure 3). The first is a welcoming scene, designed to onboard users to both the VR equipment and the overall experience. In this scene, users are guided by a voice-over, accompanied by visual cues placed throughout the environment. They are encouraged to practice basic VR interactions, such as teleporting and interacting with virtual objects and appliances. The remaining three scenes present alternative kitchen

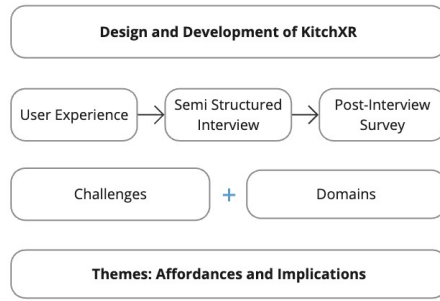


Figure 2: Schema of the methodology. Overview of the study process from prototype development to data collection and thematic analysis.

layouts, which users can switch between by interacting (grabbing or touching) with miniature models representing each scene. These three layouts—a compact kitchen, a large kitchen, and an American-style kitchen connected to a living room—were designed to offer designers a range of spatial experiences and user scenarios to explore. Each kitchen includes a variety of functioning appliance models with basic, natural interaction possibilities (Figure 4). These include a tea maker, coffee machine, toaster, robot vacuum cleaner, refrigerator, stove, and oven. On the countertops, we also placed food packages and fruits, which users can interact with freely. Only the third kitchen—the American-style layout—includes enhanced interactive features; a set of 2D UI menus, predefined tasks, and a laser pointer integrated into the user’s right hand to enable interaction with the UI. Through these elements, users can toggle between various visualization functions for selected appliances, such as viewing an exploded technical diagram of a washing machine, changing appliance models and colors, and seeing inside the devices. These menus (Figure 4) are positioned near the corresponding appliances—refrigerator, hood, and dishwasher—and operate semi-dynamically through custom Unity scripts.

3.2 Expert Evaluation Study

We used the VR application described above, KitchXR, to explore with designers the potential of employing VR for evaluating kitchen applications. We conducted an expert evaluation study with 17 participants. As noted by Lanius et al. expert review is widely adopted in UX practice, being ranked as the most common research method and the third most common evaluation method among practitioners [28]. Building on this recognition of its relevance, we complemented the evaluation with follow-up interviews and a post-experience survey. This study has been exempted from ethical approval by the Research Ethics Committee (Ref. No: 2025.362.IRB3.135), as it involves an expert evaluation.

3.2.1 Participants. 17 professionals from different departments at the headquarters of a global home appliance company in Türkiye, were purposefully recruited through an expertise sampling method. We reached participants through an open call within the company, targeting various departments involved in product development and design processes. We conducted the study in a dedicated office

space at the company. Participants’ age ranged from 22 to 48 (median 29). Only two of the participants have backgrounds in business administration while the others (15 out of 17) held a bachelor’s or higher degree in product design or visual media arts. Their professional experience ranged 1 to 24 years (median of 7 years); notably, 7 participants had less than 4 years of experience. As shown in Table 1, 8 participants held roles within UX/UI design departments, while the remaining 9 were involved in industrial design, consumer insights, R&D, or usability testing. Most participants (10 out of 17) indicated that this was their first experience using immersive VR.

Table 1: Table of participants. SKA: Small Kitchen Appliances

Part.	Age	Gender	Profession and Department	Exp. (Year)	First VR
P1	37	M	Technology Lead - Usability	12	No
P2	48	M	Global Consumer Insights	24	Yes
P3	27	F	UX Designer	3	Yes
P4	40	F	Lead Prod. Designer – SKA	16	Yes
P5	40	M	Lead Prod. Designer - Cooling	21	Yes
P6	39	M	Senior Product Designer – Cooking	15	Yes
P7	23	F	UI/UX Designer (UX/UI)	2	Yes
P8	36	F	Color/Material/Finish Designer	15	No
P9	25	F	Assistant UX/UI Designer	1	Yes
P10	25	F	Product Designer	3	No
P11	23	F	Project Assistant	2	Yes
P12	22	F	UX/UI Design Assistant	1	Yes
P13	29	M	Product Designer - SKA	7	Yes
P14	26	M	UI/UX Designer	6	No
P15	39	M	UI/UX Designer	16	No
P16	29	F	UX/UI Designer	1	No
P17	31	M	Product Designer, SKA	10	No

3.2.2 Expert Evaluation Study Procedure. The expert evaluation study began with a short briefing session explaining the VR Kitchen application and how to use the controllers. After the briefing, participants wore the headset and started freely exploring the VR Kitchen. The first scene was designed specifically to onboard participants with little or no VR experience. During this onboarding phase, users were guided to perform basic interactions such as teleporting, grabbing objects, and opening doors. Once they demonstrated familiarity with these actions, they were invited to explore three different kitchen scenes, each offering distinct spatial layouts. Participants navigated these environments freely—either by walking or teleporting—and interacted with various objects in all four scenes. During this phase, they were encouraged to think aloud, sharing their immediate impressions, emotional responses, and reflections on the visual and interactive aspects of the experience. After this free exploration, participants were guided via external voice instructions to complete four specific tasks in the third kitchen scene. These tasks were: (1) customizing the refrigerator, (2) placing packages into shelves and cupboards, (3) turning the hood on and off, (4) filling the tea maker with water from the faucet. These tasks were designed to demonstrate how VR can support scenario-based UX research. They allowed designers to experience how user behaviors can be observed during goal-driven interactions, how 2D UI menus can be operated with a laser pointer, and how differences in appliance size or interface design may influence user experience.

Semi-Structured Interview. Following the VR experience, each participant engaged in a semi-structured interview guided by 14 core questions (see Table 2). The conversational format allowed us to probe deeper into participants’ professional perspectives and

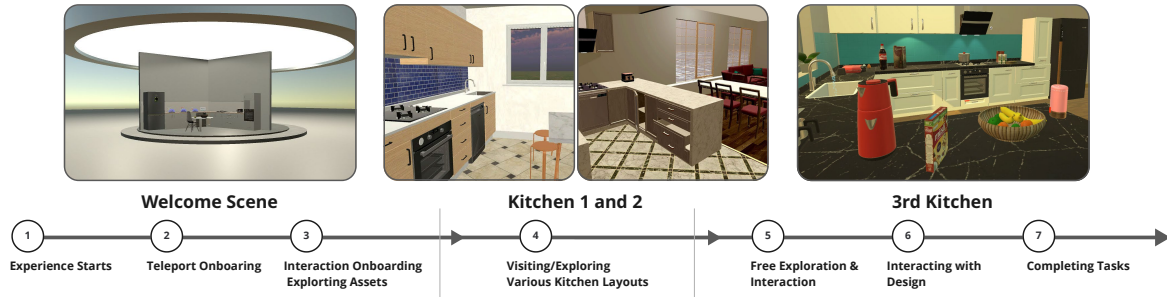


Figure 3: User journey map of the VR Kitchen experience, outlining key stages from onboarding to exploratory interaction and task completion across three virtual kitchen layouts.

explore emergent themes relevant to XR integration in home appliance design processes. Interview questions focused on three areas: (1) participants' current UX research practices and challenges in home appliance design, (2) their experiential evaluation of the VR Kitchen prototype compared to existing research tools and real kitchen environments, and (3) their envisioned applications of XR technology for user insight collection—including product conceptualizing, interface testing, and iterative feedback gathering. Specific probes examined the design effectiveness of the VR interface, the role of task scenarios in eliciting authentic user behaviors, and participants' professional assessment of how virtual prototype fidelity and interactivity levels might impact the quality of user insights collected through such tools. All interviews were audio-recorded and transcribed with the informed consent of participants. To complement our qualitative findings, we administered two standardized usability measures: the System Usability Scale (SUS) [24] and the After Scenario Questionnaire (ASQ) [29].

System Usability Scale (SUS). We used the SUS to assess the overall perceived usability of the KitchXR. The SUS is a widely recognized 10-item scale that provides a quick and reliable measure of system usability through a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). The SUS allowed us to examine whether the prototype achieved a level of usability that would enable designers to focus on their UX research practices and its objectives rather than overcoming interface barriers.

After Scenario Questionnaire (ASQ). To evaluate task experience, we employed a 3-item ASQ measuring perceptions of task difficulty, time adequacy, and system feedback on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree) [29]. The ASQ offered more granular insights than the SUS, particularly on whether task scenarios supported meaningful interactions by evaluating ease, convenience, and guidance. This is critical, as scenario design directly affects the quality of user insights in XR-based research tools. These quantitative benchmarks indicated whether the prototype met usability thresholds for professional insight collection, clarifying which aspects aligned with designer expectations and which required improvement. Complementing this, our qualitative analysis examined how XR could reshape UX research practices and identified the most valuable capabilities from the perspective of home appliance design professionals.

Table 2: Questions of semi-structured interview.

Interview Questions	Context
1 Have you experienced a VR application before? What kind of application did you try? Could you compare this experience with Virtual Kitchen?	Experience
2 Which prototyping methods do you find more efficient in the design process? Could you compare your VR experience with other prototyping methods?	Experience
3 Would conducting user experience studies and usability tests in a VR environment contribute to the process? What kind of contributions would it make?	Experience - Goal
4 Considering your current task, if you had such a tool, how frequently would you use it in the design process?	Experience - Scenario
5 What information would you like to gather from users while collecting insights in a VR experience?	Experience - Scenario
6 What kind of interactions and features could be added to make the tests more efficient?	Experience - Scenario
7 How could the kitchen product experience be enhanced with the addition of other types of functions to the Virtual Kitchen?	Experience - Func.
8 How would you prefer to ask questions to the user during the experience?	UI
9 When evaluating the overall appearance of the Virtual Kitchen, how did you find the size of the environment, the number/placement of products, the space allocated to you, etc.?	Model
10 Was the ambient light sufficient and realistic in the kitchens you experienced in the VR environment? Would you need different lighting to try out products?	Model
11 Would experiencing products in different types of kitchens contribute to the design process?	Experience
12 Were you able to interact easily with the products in the Virtual Kitchen? For example, how was opening the refrigerator door, moving products, in terms of the functions you used? How could it be improved?	Interaction
13 What are your thoughts on using a laser pointer?	Interaction
14 What are your thoughts on AR menu design? How were the choices in terms of comprehensibility and applicability?	UI

3.3 Data Analysis

In our analysis, we complimented qualitative thematic analysis of interview responses with quantitative assessment of usability metrics. This dual approach allowed us to evaluate both the practical usability of our VR prototype and gain deeper insights into designers' experiences, needs, and expectations regarding the use of VR tools for UX research.

3.3.1 Qualitative Data Analysis. The qualitative analysis was conducted using the six-phase approach to thematic analysis proposed by Braun and Clarke [7], which is well-suited for exploratory research aimed at examining user experiences and professional practices in order to identify patterns within qualitative data. Two researchers independently reviewed the dataset, noting observations and identifying potential patterns. We employed an inductive approach for systematic coding, allowing the codes to emerge directly. Codes were generated based on participants' current UX research practices, their experiences and reflections on the VR prototype,

and their suggestions for potential improvements or additional features. The final analysis identified two main problem areas in existing practices, directly addressing the first research question (RQ1), and three key themes regarding the potential of VR to support user data collection and facilitate its integration into existing design processes, addressing the second research question (RQ2). This structure allowed us to clearly delineate between the current challenges faced by designers in UX research and the envisioned future roles of XR technologies. The thematic analysis revealed three major themes corresponding to design features that could be integrated into VR tools for UX research: (1) enriching user experience design with complex scenarios and alternative testing environments, (2) integration of bio-sensory measures to enable richer, multi-modal data collection, and (3) enabling revisitation of the design process by capturing data that can inform future iterations. Each theme was supported by sub-themes that reflected how participants envisioned VR integration into their workflows. In our reporting, we also present the frequency of similar responses among participants ($n/N = 17$), offering insight into the consistency and variability of perspectives within our study cohort.

3.3.2 Quantitative Data Analysis. SUS scores were calculated following standard protocols [24]. This recoding produced values on a 0–4 scale, with 4 representing the most positive response. The recoded scores were summed and multiplied by 2.5 to yield a final score on a 0–100 scale. Scores above 68 generally indicate above-average usability [24]. In our study, individual participant scores ranged from 47.5 to 82.5 (mean of 62.5). While the overall SUS score summarizes general usability, Table 3 presents the mean raw Likert ratings (1–5) for each item, highlighting specific aspects of the VR Kitchen that participants rated more positively or negatively and providing insight into usability strengths and areas for improvement. ASQ responses were analyzed to examine task-specific interactions participants engaged in the VR kitchen experience. For each of the three ASQ items, we report the frequency of participant ratings on the 1–7 scale, providing a descriptive summary of responses (Figure 5). These quantitative measures from the SUS and ASQ complemented our thematic analysis, contributing to the discussion on practicality of use under the discussion on commonly targeted in appliance UX research user insight domains (SUS), and to the theme of complex scenarios and alternative testing environments, where we discuss participants' responses to the ASQ.

4 FINDINGS

This section presents findings from our mixed-methods expert evaluation study assessing the utility and perceived value of the VR-based prototype KitchXR for collecting user insights in home appliance design. The study addressed two questions: (RQ1) What challenges and limitations exist in current UX research workflows for home appliances? and (RQ2) How might immersive VR extend or enhance these workflows for iterative testing?

The findings are organized into three main parts. First, we examine current UX research practices in product design and the emerging opportunities for VR integration, identifying three key limitations: access, timing, and workflow integration. Second, we analyze user insight domains commonly targeted in appliance UX research—practicality, aesthetics, and emotional connection—and

how VR can support them. Third, we present three overarching themes derived from qualitative analysis: (1) enriching user experience through complex and contextual scenarios, (2) integration of biosensory measures for deeper insight, and (3) enabling revisitation of the design process across multiple stages. Together, these findings provide a nuanced understanding of where immersive VR tools may meaningfully augment existing design research practices.

4.1 Opportunities and Challenges for Integration

Participants, working for the same global home appliance company, described their current UX research practices as geographically distributed across national and international locations. These practices typically involve using 3D-printed or foam-core prototypes at full scale in dedicated testing studios. UI interactions are often simulated through embedded tablets or smartphones, enabling users to engage with interfaces as if integrated into the physical product. During sessions, facilitators guide participants through predefined scenarios, while UX researchers observe behaviors, collect verbal feedback, and sometimes record physiological data such as eye tracking or skin conductance. Our analysis identified three critical limitations within this workflow that highlight the potential value and challenges of immersive VR tools: (1) restricted access to geographically diverse and demographically varied user data, (2) time constraints that delay or limit user feedback in early development stages, and (3) challenges in integrating VR into existing digital design and engineering pipelines.

4.1.1 Access: Expanding User Data Collection. The first challenge centers on geographical and logistical constraints. Participants (10/17) described the difficulty of transporting large physical prototypes across locations and the complications involved in setting up consistent testing environments abroad. Many believed that VR could allow UX teams to conduct more frequent and remotely distributed studies, increasing both the variety and number of users involved (12/17), while enabling faster data collection (10/17). Several participants (7/17) noted that such studies could also be more easily automated in VR environments, offering the possibility of collecting rich data asynchronously and at scale. A few participants (2/17) emphasized the added benefit of standardizing study conditions across locations via controlled VR environments. Participants also pointed out that VR could better simulate product variants and materials that are difficult to prototype physically. For example, 3D-printed models are usually monochrome and made of materials that differ significantly from the final product. VR could address this by allowing users to preview different appliance models, color variations, and finish types more accurately (14/17). Moreover, immersive environments can simulate diverse kitchen settings (6/17), and even replicate individual users' home environments (2/17), offering more contextually relevant scenarios for evaluation.

4.1.2 Timing: Enabling Earlier and More Frequent Feedback. The second issue relates to the temporal distance between product prototyping and user testing. Participants (14/17) reported that current processes often delay user testing until late in the development cycle, limiting the ability to incorporate feedback into the final product. Three participants explicitly mentioned that they are not



Figure 4: Screenshots from the application. Left: Oven is interactible in exploded-perspective mode. Middle: Tea-maker as a small home appliance. Right: 2D Menu presents options.

involved in the final stages of the design process, and therefore cannot respond to final usability issues or design revisions. Participants (9/17) suggested that VR could help fill this temporal gap by enabling earlier and more frequent user testing sessions. They emphasized that integrating VR into the prototyping process could foster stronger collaboration across teams and ensure that changes based on user feedback are validated more quickly. Some participants (7/17) highlighted VR's potential in collecting insights even in the concept development phase (2/17), helping designers *"better grasp how the finished version looks and works"* (P17), and potentially improving design quality through earlier validation loops.

4.1.3 Integration Challenges. Despite its potential, the reason of not adapting VR into existing workflows, mostly relies on integration challenges identified by our participants. One significant barrier that we also confront in our study, lies in the incompatibility between design software formats. Product designers and engineers commonly use CAD tools which are based on NURBS (Non-Uniform Rational B-Splines) to present 3D models with high geometric precision. By contrast, VR environments and game engines such as Unity primarily rely on polygon-based 3D models, necessitating an often time-consuming conversion process to adapt CAD-based product model for use in immersive experiences (P6). This conversion introduces friction into fast-paced design cycles and may deter adoption without automated translation pipelines. Additionally, participants noted that virtual environments still fall short in delivering certain physical feedback cues—particularly those related to ergonomics and haptics—which are critical in evaluating physical appliances (8/17). This limitation was especially noted by industrial designers, who emphasized the continued need for physical prototyping when evaluating aspects such as button resistance, weight, or tactile feedback.

Nonetheless, most participants viewed VR not as a replacement but as a valuable addition to their existing UX research toolkit. It was considered a complementary method for expanding the scope of insight collection, particularly in areas where physical prototyping is difficult, expensive, or logistically impractical. As one UX/UI designer (P2) put it: *"As long as you expand the toolbox, you will have new areas that you need that tool for."*

4.2 User Insight Domains in UX Research

Analysis revealed three primary areas that participants address in UX research can be meaningful in VR: (1) practicality of use, (2) material quality and aesthetics, and (3) connection between user

and product. UX designers focused primarily on user experience when interacting with different features, while product designers emphasized physical qualities such as ergonomics, material, and color.

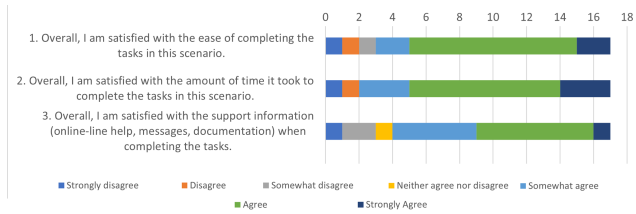
4.2.1 Practicality of Use. Participants defined practicality in terms of ease of use and functionality, emphasizing the first encounter phase and initial learning process (2/17). Both UX/UI designers (P2, P12) and product designers (P10) identified the learning phase—when users are first introduced to the product and learn to use its UI or features—as crucial for UX research. As one participant explained: *"When you are learning how to use a tool for the first time, it takes some effort and time to adopt, and then it gets automated. This phase gives us crucial insights into how to guide the user experience in a way it is more practical to use the product"* (P2). The learning curve provides essential information about user experience and product design. Questions sought by designers in this domain include: *"Can the user feel comfortable and confident while using the product? How will they experience the first learning process, where will they press, will they press accurately?"* (P12). Two participants (P2, P10) suggested that instructing users on product operation while observing their performance, as well as understanding how users naturally operate the product, provides valuable information that could be implemented in VR applications.

The System Usability Scale (SUS) yielded a score of 62.3, below the average benchmark of 68. Table 3 summarizes mean scores for each item. Interviews indicated that lower scores were largely attributable to participants' limited VR experience, which made adapting to the controllers and virtual environment dynamics challenging. Strengths included low perceived complexity ($M = 2.14$ for unnecessary complexity, inconsistency, and cumbersomeness) and strong functional integration ($M = 3.57$), suggesting that the prototype's features felt cohesive and conceptually clear. In contrast, middling ratings for ease of use, confidence, and learnability ($M = 2.85$ across related items) highlight friction in onboarding and first-time use. Addressing these challenges—particularly through enhanced guidance, feedback, and control affordances—could substantially improve both user confidence and overall usability.

4.2.2 Material Quality and Aesthetics. Participants (11/17) focused on how users respond to physical properties and visual attributes such as material type, weight, volume, and visual design. They noted that current rapid prototyping methods have limitations: foam prototypes are faster to produce but lack detail in visual and material

Table 3: System Usability Scale (SUS) responses of 17 participants. 1-Strongly Disagree / 5-Strongly Agree

SUS Items	Mean
1. I think that I would like to use this system frequently.	3.57
2. I found the system unnecessarily complex.	2.14
3. I thought the system was easy to use.	2.85
4. I think that I would need the support of a technical person to be able to use this system.	2.85
5. I found that the various functions in this system were well integrated.	3.57
6. I thought that there was too much inconsistency in this system.	2.14
7. I would imagine that most people would learn to use this system very quickly.	2.85
8. I found the system very cumbersome to use.	2.14
9. I felt very confident using the system.	2.85
10. I needed to learn a lot of things before I could get going with this system.	2.14

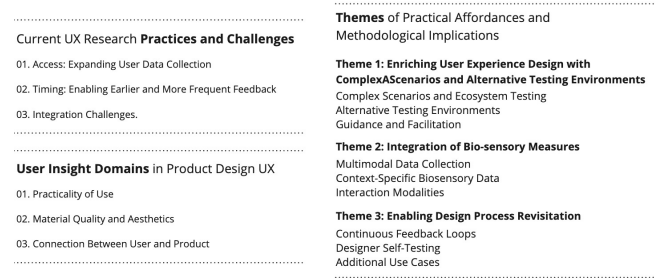
**Figure 5: After Scenario Questionnaire (ASQ) responses of 17 participants.**

quality, providing only basic information about form, mechanics, and ergonomics (4/17). Virtual prototypes such as rendered images present visual details but lack functional characteristics (5/17). One product designer (P8) explained how VR could support insights into user preferences and aesthetic perception: *"I would like to learn about users' love and hate perception"* regarding various models presented with visual detail alongside functional features.

4.2.3 Connection Between User and Product. The third domain concerns the emotional bond users develop with products (3/17). One UX designer (P14) emphasized that establishing emotional bonds with physical products requires users to *"see, touch, and feel"* the product, highlighting the importance of embodied interaction. Another participant (P2) noted that this connection is influenced by ease of use, requiring intuitive interaction that aligns with users' previous experiences with similar products. The connection domain appears to be shaped by the other two domains—practicality of use and material quality and aesthetics—which together provide essential information about user-product relationships.

4.3 Themes of Practical Affordances and Methodological Implications

Building on the previously identified challenges in current UX research practices and the user insight domains observed across design roles, this subsection presents three themes that emerged from our qualitative analysis (see Fig. 6). These themes illuminate how immersive VR tools may enhance user insight generation in product design workflows. Each theme corresponds to a distinct area where VR introduces new capacities or reconfigures existing ones: (1) enriching user experience design through complex scenarios and flexible testing environments, (2) integrating biosensory measures for deeper behavioral and emotional analysis, and

**Figure 6: Overview of key findings from the expert evaluation study. Left: Current UX research practices and user insight domains in product design; Right: 3 thematic areas that highlight the practical affordances and methodological implications of integrating VR into UX research.**

(3) enabling iterative revisitation and reflection during early-stage design. The findings highlight both the practical affordances and methodological implications of embedding immersive tools in the UX research process.

4.3.1 Theme 1: Enriching User Experience Design with Complex Scenarios and Alternative Testing Environments. Participants gave high ratings on the After Scenario Questionnaire (ASQ) for evaluating tasks, scenarios, and activities in the VR experience (Fig. 5). Across the three ASQ items—ease of task completion, time required, and adequacy of system guidance—12 of the 17 participants responded *"somewhat agree"* or higher, indicating that most found the tasks straightforward and well supported. However, while task execution was perceived as easy from an interaction standpoint, participants questioned the depth of user engagement the current scenarios could support. They recommended incorporating more complex, goal-oriented sequences (e.g., preparing breakfast) that better reflect everyday kitchen activities (4/17) and ensuring functional interactions with model appliances match real-world expectations (3/17). As one participant (P3) explained: *"I want to learn how many modes the user is employing in real life. We know that users don't employ all the 14 different modes in their washing machine. But the question is, how many modes they are using—2 or 4? And what if they only see these 4 modes instead of 14? How would they react? We cannot get this kind of insights since we are not able to produce many alternatives of a physical prototype."*

Complex Scenarios and Ecosystem Testing. UX designers sought sophisticated user insights in complex situations, such as using multiple connected devices, sometimes including mobile devices (4/17). Participants suggested that VR provides a suitable environment for connecting different prototypes in working relationships within larger use contexts. One participant (P16) noted: *"I would like to see how you feel in such an environment as the washing machine rang, buy clothes, the tea maker rang, the doorbell rang, etc. We always look at the products individually, but when they come together and create an ecosystem."*

Alternative Testing Environments. Based on their VR Kitchen experience, participants suggested novel ways of designing test

environments and interaction possibilities (3/17). One UX/UI designer (P15) proposed that unlike real-life testing with mock-up kitchens, VR could create environments exclusively for specific cases, allocating more time for product-specific tasks and scenarios: *"For some products, it may be more accurate to see the prototype alone, not in the environment. There are specific tasks related to that product, especially on the UI/UX side, where the environment the user is in does not have much effect."* Another participant (P7), a product designer, agreed that using alternative user interaction techniques for situations where it is unnecessary to build an environment testing setting for all models, but capture user interaction solely on one model with specific and detailed user insights. One participant (P5) gave positive feedback on the functionality where the VR-kitchen application displayed the internal working, such as air movement in a fridge model, and shared that these types of interactions, impossible to convey with physical prototypes, can be used as a general functionality in the VR application. P1 (UX) suggested that VR could enrich the user experience by adding more product-specific use cases to get more user feedback.

Guidance and Facilitation. Participants (10/17) emphasized the need for guidance in VR user experience studies, similar to existing protocols where UX researchers are present during testing. This could involve UX researchers being virtually and/or physically present, either embodied as avatars (4/17) or through voice-based interaction without visual representation (4/17). Participants (3/17) suggested incorporating visual, auditory, and spatial cues to help users navigate through steps and tasks.

4.3.2 Theme 2: Integration of Bio-sensory Measures. Participants (9/17) endorsed VR's potential to provide broader user observation capabilities through in-situ assessments of different user experience aspects. While bio-tracking techniques like eye-tracking are already used in user insight studies (3/17), they are not collected systematically or automatically (1/17). Embedded bio-tracking sensors in VR headsets (6/17) and aligned study scenarios (3/17) could address deeper and broader user insight requirements.

Multimodal Data Collection. More in-depth user experience information could be supported through physiological measures (heart rate, galvanic skin response) (4/17), motion tracking (6/17), and eye tracking (8/17). One participant (P1) highlighted the advantages of coupling eye-tracking and movement tracking: *"What the user is looking at and how the user views the interaction is important for us. Also, the movement; what kind of movements does the user make when using the prototype. This can be recorded and analyzed to give better insight into the process of interaction."* Another participant (p12) suggested in understanding how users interact with the UI features, *"the hand movements are important. And it's good to know how you feel"*. Eye and hand tracking coupled is mentioned by three participants (p12, p16, p11) while the finger sensitivity put forwards as a necessary data to complement insights for small appliances (p12).

Context-Specific Biosensory Data. Participants suggested that different biosensory data should be selected according to specific insights being collected. For instance they suggested that skin conductance and heart rate might be suitable for connection insights

(P2), eye-tracking coupled with hand-tracking for practicality insights (P10, P12), and eye-tracking coupled with skin conductance for aesthetic insights (P8).

Interaction Modalities. Three participants found that controller-based interaction broke the feeling of natural manual interaction, especially with small appliances having multiple parts. Hand gloves with vibro-tactile feedback were recommended as better alternatives to standard VR controllers (P15). This was less problematic for UI feature interactions (3/17). One participant (P2) suggested VR could be particularly suitable for UI research *"as we are not concerned with measuring finger sensitivity or other ergonomic issues there. We are interested in the flow of menus, whether it follows a certain logic and it is understandable."*

4.3.3 Theme 3: Enabling Design Process Revisitation. Participants suggested that VR can be used in early design stages (P11, P5) to support designers' decision-making processes (P17, P11). The early concept development phase could particularly benefit from VR-collected user data to establish design directions (3/17). One UI designer working on novel interface design and interaction techniques (P12) noted it would benefit stages where imagination is primarily used.

Continuous Feedback Loops. Different design alternatives can be tested across various design stages, providing continuous feedback loops (P16, P17). One UX designer suggested VR could be useful for preparing user experience scenarios for formal physical tests (P15) and for pilot testing to identify potential problems and revise scenarios before formal physical studies (P12, P15).

Designer Self-Testing. Five participants—two UX/UI designers (P16, P12) and three product designers (P7, P11, P17)—suggested VR could enable designers to experience prototypes themselves and foresee problems typically found later in UX testing. As one participant (P11) noted: *"The biggest advantage would be to try it ourselves. We could have found the bugs that will be found in some future tests early and here. In fact, we could perform these tests ourselves."* VR could also bring together teams of designers and engineers to test prototypes collaboratively (P8, P14).

Additional Use Cases. Participants identified additional potential application areas: remote collaboration (2/17), decision-making (4/17), co-design (5/17), prototyping across various design and development stages (6/17), scenario-testing for UX studies (2/17), benchmarking and market research (2/17), training and professional development (1/17), ergonomic research (1/17), and color material finish (CMF) design (2/17).

5 DISCUSSION

This study explored two interrelated research questions: (RQ1) What challenges and limitations exist in current UX research workflows for home appliances? and (RQ2) How might immersive VR extend or enhance these workflows for iterative testing? The discussion below engages with these questions by synthesizing our findings with relevant literature in product design, UX, HCI, Spatial Computing, and XR.

Across product design industries such as home appliances, UX research traditionally relies on physical prototyping—yet participants revealed persistent frictions in these practices. These include logistical constraints in transporting large or fragile models, difficulties replicating contextual qualities, and the late-stage timing of UX tests. Our findings echo prior work on inefficiencies in physical prototyping workflows, especially in globally distributed teams and high-cost industries [8, 11, 13]. These challenges inform the growing interest in alternative research environments. While Living Lab methodologies enable in-situ testing, they introduce complexity in setup, lack standardization, and often fail to meet industry demands for rapid, repeatable, low-cost methods. In contrast, immersive VR systems like our KitchXR prototype offer scalable, remote, and controlled UX evaluations [4, 42]. This study contributes by examining how immersive tools might realistically be integrated into existing design workflows. Rather than focusing on end-user acceptance, we foreground the perspectives of design professionals—UX/UI designers, product designers, and engineers—revealing how they assess the value, limitations, and applicability of VR in their own practices. This practitioner-centered view remains underexplored in XR adoption research. However, it also limits our findings to projected use rather than observed long-term integration.

Participants emphasized VR's ability to simulate richer scenarios and alternative contexts. This supports Bruno et al.'s argument that realistic context fidelity is key for meaningful UX insights [8]. However, several participant suggestions—such as integrating haptic gloves—require critical consideration. While richer interaction fidelity could enhance realism, these technologies also risk reducing accessibility and scalability, undermining one of VR's key advantages: low-cost, distributed testing. Thus, design teams must weigh realism against inclusivity and workflow feasibility when envisioning adoption. Another opportunity lies in VR's multimodal data collection. Eye tracking, motion tracking, and biosensory measures (e.g., galvanic skin response) embedded in headsets can deepen understanding of user emotion, cognition, and behavior. These findings build on work by Krauß et al. [26], though participants stressed the importance of aligning data types with research goals—e.g., combining hand tracking with motion data for ergonomics, or eye tracking with skin conductance for aesthetic evaluation. This suggests that while VR affords richer data capture, its true methodological contribution lies in selectively integrating these measures where they extend—not duplicate—existing UX research practices. Participants also envisioned VR supporting early-stage design work. Designers described using immersive environments for self-testing, bug detection, and rapid prototyping—resonating with speculative and generative XR practices advocated by Höggenmüller et al. [22]. VR becomes not just a testing platform but a creative and reflexive tool in iterative workflows. The semi-structured interviews show that KitchXR functioned not only as an evaluation setup but also as a probe for eliciting designers' expectations. Immersive VR also enables asynchronous collaboration among globally distributed teams. Echoing prior studies [11, 13], participants imagined Metaverse-style environments facilitating UX research and coordination when physical access to users or teammates is constrained.

Despite these benefits, adoption barriers remain. Participants noted the friction between high-fidelity CAD models and game engine formats, echoing Deininger et al. [14]. Time-consuming mesh

conversions hinder rapid iteration. Concerns also surfaced about the lack of tactile realism and ergonomic fidelity—particularly for evaluating physical properties crucial to product design. These limitations confirm that VR is best positioned as an augmenting layer, complementing physical prototyping and traditional UX research methods, rather than replacing them.

Together, these insights contribute to HCI and design by mapping how spatial computing and immersive VR might integrate into product development processes. We offer an empirically grounded view of how design teams imagine the affordances and limits of VR in practice. Our study also contributes to XR literature by identifying domain-specific insight domains—such as practicality, material perception, and emotional connection—that immersive tools like KitchXR can uniquely support. While our study focused on kitchen appliances and a single prototype, the findings have broader relevance across product domains where spatial interaction, physical form, and context are key. In relation to RQ2, KitchXR illustrates how immersive environments can act both as research instruments and as generative probes, shaping the types of insights available to design professionals. Yet their adoption depends not only on technical advances but on organizational readiness and willingness to reshape workflows—an ongoing challenge for future research.

6 CONCLUSION

This study examined how immersive VR tools might be integrated into UX research within the product design processes of home appliance industries. We developed KitchXR, a scenario-based VR kitchen prototype, and conducted expert evaluation study with 17 professionals from a global home appliance company. Through interviews and usability evaluations, we identified key workflow challenges and explored the design affordances VR can offer for user-centered research. The original contribution of this study lies in foregrounding the perspectives of product team members—UX/UI designers, product designers, and decision-makers—on how immersive tools could realistically fit into their existing practices. Rather than focusing on end-user interaction, our approach centers on how design professionals evaluate the potential, limitations, and adoption of XR tools in early-stage development.

Our findings show that current UX practices, heavily reliant on physical prototyping, face constraints in access, timing, and iteration. KitchXR was perceived as a complementary tool that enables remote, immersive, and scenario-rich user testing. We identified three key affordances for XR-based UX tools: enabling complex, contextual scenarios; supporting biosensory integration for deeper insight; and fostering iterative, revisitable design processes. These address RQ1 by highlighting persistent gaps in current UX research practices and RQ2 by illustrating how VR can extend methodological options to fill them. Beyond the immediate case of kitchen appliances, these findings have broader implications for design and HCI. They suggest that VR's role in UX research is not merely to replicate physical prototyping but to create new conditions for insight generation—especially where context, iteration, and multimodality matter. For industry, this points toward a shift in how design teams conceive user research infrastructures. For research, it underscores the value of VR as both a methodological probe and a platform for integrating qualitative and biosensory data.

At the same time, the study's contributions are bounded. Its original focus on speculative perspectives—rather than long-term deployments—limits claims about sustained integration. Additional limitations include the domain-specific context and reliance on a single VR prototype. Future work should examine longitudinal use across full design cycles, investigate integration of tactile and ergonomic evaluation, and develop frameworks for merging immersive prototyping with engineering pipelines. By doing so, we can better understand not only how VR responds to today's UX research challenges but also how it may reshape the methodological landscape of design research in the future.

ACKNOWLEDGMENTS

This study was designed as an expert evaluation and was approved and exempted by the ethics committee upon submission, under reference number 2025.362.IRB3.135. The authors gratefully acknowledge Serra Cılızoglul for her valuable contribution to the development of the virtual kitchens. This work was carried out within the scope of the TÜBİTAK 2244 project (Project No. 118C095) and was sponsored by Beko.

REFERENCES

- [1] G ; Alce, K ; Hermodsson, M ; Wallergård, L ; Thern, and T Hadzovic. 2015. A Prototyping Method to Simulate Wearable Augmented Reality Interaction in a Virtual Environment-A Pilot Study. *International Journal of Virtual Worlds and Human Computer Interaction* 3 (2015), 2368–6103. doi:10.11159/vwhci.2015.003
- [2] Susanna Aromaa and Kaisa Väänänen. 2016. Suitability of virtual prototypes to support human factors/ergonomics evaluation during the design. *Applied Ergonomics* 56 (2016), 11–18. doi:10.1016/j.apergo.2016.02.015
- [3] Leif P. Berg and Judy M. Vance. 2017. An Industry Case Study: Investigating Early Design Decision Making in Virtual Reality. *Journal of Computing and Information Science in Engineering* 17, 1 (mar 2017). doi:10.1115/1.4034267/472843
- [4] Leif P. Berg and Judy M. Vance. 2017. Industry use of virtual reality in product design and manufacturing: a survey. *Virtual Reality* 21, 1 (mar 2017), 1–17. doi:10.1007/S10055-016-0293-9/TABLES/9
- [5] M Bordegoni, U Cugini, G Caruso International Journal on ..., and undefined 2009. 2009. Mixed prototyping for product assessment: a reference framework. *Springer* 3 (2009), 177–187. Issue 3. doi:10.1007/s12008-009-0073-9
- [6] Monica Bordegoni and Francesco Ferrise. 2013. Designing interaction with consumer products in a multisensory virtual reality environment. <https://doi.org/10.1080/17452759.2012.762612> 8 (3 2013), 51–64. Issue 1. doi:10.1080/17452759.2012.762612
- [7] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [8] Fabio Bruno and Maurizio Muzzupappa. 2010. Product interface design: A participatory approach based on virtual reality. *International Journal of Human Computer Studies* 68, 5 (may 2010), 254–269. doi:10.1016/j.ijhcs.2009.12.004
- [9] Alisa Burova, John Mäkelä, Jaakko Hakulinen, Tuuli Keskinen, Hanna Heinonen, Sanni Siltanen, and Markku Turunen. 2020. Utilizing VR and Gaze Tracking to Develop AR Solutions for Industrial Maintenance. In *Conference on Human Factors in Computing Systems - Proceedings*, Vol. 20. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376405
- [10] Gaetano Cascini, Jamie O'Hare, Elies Dekoninck, Niccolò Becattini, Jean François Boujut, Fatma Ben Guefrache, Iacopo Carli, Giandomenico Caruso, Lorenzo Giunta, and Federico Morosi. 2020. Exploring the use of AR technology for co-creative product and packaging design. *Computers in Industry* 123 (2020). doi:10.1016/j.compind.2020.103308
- [11] Senthil K. Chandrasegaran, Karthik Ramani, Ram D. Sriram, Imre Horváth, Alain Bernard, Rami F. Harik, and Wei Gao. 2013. The evolution, challenges, and future of knowledge representation in product design systems. *Computer-Aided Design* 45, 2 (feb 2013), 204–228. doi:10.1016/j.CAD.2012.08.006
- [12] Joshua Q. Coburn, Ian Freeman, and John L. Salmon. 2017. A Review of the Capabilities of Current Low-Cost Virtual Reality Technology and Its Potential to Enhance the Design Process. doi:10.1115/1.4036921
- [13] Marianna J. Couliantanos, Ilka Rodriguez-Calero, Shanna R. Daly, and Kathleen H. Sienko. 2020. Stakeholder Engagement With Prototypes During Front-End Medical Device Design: Who Is Engaged With What Prototype? *Frontiers in Biomedical Devices, BIOMED - 2020 Design of Medical Devices Conference, DMD 2020* (jul 2020). doi:10.1115/DMD2020-9020
- [14] Michael Deininger, Shanna R. Daly, Jennifer C. Lee, Colleen M. Seifert, and Kathleen H. Sienko. 2019. Prototyping for context: exploring stakeholder feedback based on prototype type, stakeholder group and question type. *Research in Engineering Design* 30, 4 (oct 2019), 453–471. doi:10.1007/S00163-019-00317-5/FIGURES/5
- [15] P. Fillatreau, J. Y. Fourquet, R. Le Bolloc'H, S. Cailhol, A. Datas, and B. Puel. 2013. Using virtual reality and 3D industrial numerical models for immersive interactive checklists. *Computers in Industry* 64, 9 (2013), 1253–1262. doi:10.1016/j.compind.2013.03.018
- [16] Florin Timotei Ghiurău, Mehmet Aydin Baytas, and Casper Wickman. 2020. ARCAR: On-Road Driving in Mixed Reality by Volvo Cars. In *UIST 2020 - Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, Inc, 62–64. doi:10.1145/3379350.3416186
- [17] Lorenzo Giunta, Elies Dekoninck, James Gopsill, and Jamie O'Hare. 2018. A review of augmented reality research for design practice: Looking to the future. *Proceedings of NordDesign: Design in the Era of Digitalization, NordDesign 2018* (2018).
- [18] David Goedicke, Jami Li, Vanessa Evers, and Wendy Ju. 2018. VR-OOM: Virtual reality on-road driving simulation. In *Conference on Human Factors in Computing Systems - Proceedings*, Vol. 2018-April. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3173574.3173739
- [19] Denis Gračanin, Mohamed Handosa, and Hicham G. Elmongui. 2017. A Service Infrastructure for Human-Centered IoT-Based Smart Built Environments. In *Distributed, Ambient, and Pervasive Interactions*, Norbert Streitz and Panos Markopoulos (Eds.). Lecture Notes in Computer Science, Vol. 10291. Springer, 262–274. doi:10.1007/978-3-319-58697-7_19
- [20] Uwe Gruenefeld, Jonas Auda, Florian Mathis, Stefan Schneegass, Mohamed Khamis, Jan Gugenheimer, and Sven Mayer. 2022. Vrception: Rapid prototyping of cross-reality systems in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [21] Marc Hassenzahl. 2010. *Experience Design: Technology for All the Right Reasons*. Synthesis Lectures on Human-Centered Informatics, Vol. 3. Morgan & Claypool Publishers. 1–95 pages. doi:10.2200/S00261ED1V01Y200911HCI006
- [22] Marius Hoggemüller, Martin Tomitsch, Luke Hespanhol, Tram Thi Minh Tran, Stewart Worrall, and Eduardo Nebot. 2021. Context-based interface prototyping: Understanding the effect of prototype representation on user feedback. *Conference on Human Factors in Computing Systems - Proceedings* (may 2021). doi:10.1145/3411764.3445159
- [23] Stephen S. Intille, Kent Larson, J. S. Beaudin, J. Nawyn, E. Munguia Tapia, and P. Kaushik. 2005. A living laboratory for the design and evaluation of ubiquitous computing technologies. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems* (Portland, OR, USA) (CHI EA '05). Association for Computing Machinery, New York, NY, USA, 1941–1944. doi:10.1145/1056808.1057062
- [24] John Brooke. 1996. SUS: A 'Quick and Dirty' Usability Scale. *Usability Evaluation In Industry* (6 1996), 207–212. doi:10.1201/9781498710411-35
- [25] Pascal Knierim, Paweł W. Woźniak, Yomna Abdelrahman, and Albrecht Schmidt. 2019. Exploring the potential of augmented reality in domestic environments. *Proceedings of the 21st International Conference on Human-Computer Interaction with Mobile Devices and Services, MobileHCI 2019* (2019). doi:10.1145/3338286.3340142
- [26] Veronika Krauß, Michael Nebeling, Florian Jasche, and Alexander Boden. 2022. Elements of XR Prototyping: Characterizing the Role and Use of Prototypes in Augmented and Virtual Reality Design. *Conference on Human Factors in Computing Systems - Proceedings* (4 2022). doi:10.1145/3491102.3517714
- [27] S. F. Kuliga, T. Thrash, R. C. Dalton, and C. Hölischer. 2015. Virtual reality as an empirical research tool - Exploring user experience in a real building and a corresponding virtual model. *Computers, Environment and Urban Systems* 54 (nov 2015), 363–375. doi:10.1016/j.compenvurb.2015.09.006
- [28] Candice Lanius, Ryan Weber, and Joy Robinson. 2021. User Experience Methods in Research and Practice. *Journal of Technical Writing and Communication* 51, 4 (2021), 350–379. arXiv:https://doi.org/10.1177/00472816211044499 doi:10.1177/00472816211044499
- [29] James R. Lewis. 1991. Psychometric evaluation of an after-scenario questionnaire for computer usability studies: the ASQ. *SIGCHI Bull.* 23, 1 (Jan. 1991), 78–81. doi:10.1145/122672.122692
- [30] Brian Mac Namee, Pauline Rooney, Patrick Lindstrom, Andrew Ritchie, Frances Boylan, Greg Burke, et al. 2006. Serious Gordon: using serious games to teach food safety in the kitchen. In *Proceedings of the 9th International Conference on Computer Games: AI, Animation, Mobile, Educational & Serious Games (CGAMES06)*. Citeseer.
- [31] Michael R. Marner, Ross T. Smith, Shane R. Porter, Markus M. Broecker, Benjamin Close, and Bruce H. Thomas. 2011. Large Scale Spatial Augmented Reality for Design and Prototyping. *undefined* (2011), 231–254. doi:10.1007/978-1-4614-0064-6_10
- [32] Michael R. Marner and Bruce H. Thomas. 2014. Spatial augmented reality user interface techniques for room size modelling tasks. *Conferences in Research and Practice in Information Technology Series* 150 (2014), 39–45.

- [33] Bruno Marques, Hugo Ferreira, Sara Oliveira, and Beatriz Sousa Santos. 2024. Integrating Remote Participants into Co-Located Collaborative Tasks through VR and AR. In *Proceedings of the 2024 IEEE 3rd International Conference on Intelligent Reality (ICIR)*. IEEE, 1–4.
- [34] Mathias Müller, Tobias Günther, Dietrich Kammer, Jan Wojdziak, Sebastian Lorenz, and Rainer Groh. 2016. Smart Prototyping - Improving the Evaluation of Design Concepts Using Virtual Reality. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 9740 (2016), 47–58. doi:10.1007/978-3-319-39907-2_5
- [35] Patrick Olivier, Guangyou Xu, Andrew Monk, and Jesse Hoey. 2009. Ambient kitchen: designing situated services using a high fidelity prototyping environment. In *Proceedings of the 2nd International Conference on Pervasive Technologies Related to Assistive Environments (Corfu, Greece) (PETRA '09)*. Association for Computing Machinery, New York, NY, USA, Article 47, 7 pages. doi:10.1145/1579114.1579161
- [36] G Pahl, W Beitz, J Feldhusen, K.-H Grote, Ken Wallace, and Lucienne Blessing. [n. d.]. *Engineering Design A Systematic Approach* Third Edition. ([n. d.]).
- [37] Cristian Pampărău and Radu Daniel Vatavu. 2020. A Research Agenda Is Needed for Designing for the User Experience of Augmented and Mixed Reality: A Position Paper. *ACM International Conference Proceeding Series* (11 2020), 323–325. doi:10.1145/3428361.3432088
- [38] Fumio KISHINO Paul MILGRAM. 1994. A Taxonomy of Mixed Reality Visual Displays. *IEICE TRANSACTIONS on Information* E77-D, 12 (December 1994), 1321–1329.
- [39] Ingrid Pettersson, Mari Anne Karlsson, and Florin Timotei Ghiurau. 2019. Virtually the same experience? Learning from user experience evaluation of in-vehicle systems in VR and in the field. In *DIS 2019 - Proceedings of the 2019 ACM Designing Interactive Systems Conference*. Association for Computing Machinery, Inc, New York, NY, USA, 463–473. doi:10.1145/3322276.3322288
- [40] Mel Slater, Bernhard Spanlang, Maria V. Sanchez-Vives, and Olaf Blanke. 2010. First Person Experience of Body Transfer in Virtual Reality. *PLoS ONE* 5, 5 (may 2010), e10564. doi:10.1371/journal.pone.0010564
- [41] VITALISE Consortium. 2021. Virtual Health & Wellbeing Living Lab Infrastructure (VITALISE) Project. Horizon 2020 EU Project. <https://vitalise-project.eu/HarmonizingLivingLabproceduresacrossmultisitehealth&wellbeingenvironments>.
- [42] Peng Wang, Shusheng Zhang, Mark Billingham, Xiaoliang Bai, Weiping He, Shuxia Wang, Mengmeng Sun, and Xu Zhang. 2020. A comprehensive survey of AR/MR-based co-design in manufacturing. *Engineering with Computers* 36, 4 (2020), 1715–1738. doi:10.1007/s00366-019-00792-3
- [43] Johanna Katharina Weber, Margarita Esau-Held, Marvin R. G. Schiller, Eike Martin Thaden, Dietrich Manstetten, and Gunnar Stevens. 2023. Designing an Interaction Concept for Assisted Cooking in Smart Kitchens: Focus on Human Agency, Proactivity, and Multimodality. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS)*. 1128–1144. doi:10.1145/3563657.3595975
- [44] Josef Wolfartsberger. 2019. Analyzing the potential of Virtual Reality for engineering design review. *Automation in Construction* 104 (aug 2019), 27–37. doi:10.1016/j.autcon.2019.03.018
- [45] Yu-Yen Chung Yu-YenChung, Penn S State Health Milton Hershey, and Balakrishnan Prabhakaran. 2023. Performance and User Experience Studies of HILLES: Home-based Immersive Lower Limb Exergame System. *Proceedings of the 14th Conference on ACM Multimedia Systems* (6 2023), 62–73. doi:10.1145/3587819.3590985